

# Simulation of Epileptic Neuronal Dynamics and Treatment Intervention

## Introduction

Epilepsy is a neurological disorder characterised by abnormal, excessive neuronal firing, leading to recurrent seizure episodes. From a neuronal perspective, the main characteristic is hyper-excitability, often involving excessive synaptic connectivity, as well as other conditions, like reduced inhibition or altered membrane properties. This disorder is also associated with increased synchrony among neurons and a higher frequency of bursting activity.

To explore these dynamics computationally, a basic network of Leaky Integrate-and-Fire (LIF) neurons with sparse excitatory synaptic connections was implemented. This model allows to simulate and compare networks of neurons under three conditions: normal activity, epileptic behaviour and treated condition, whose aim is to mimic the effects of therapeutic interventions or anti-epileptic drugs.

## Goals

The main objective of this project is to create a simple computational framework, capable of simulating and comparing neuronal dynamics under the three conditions mentioned above: healthy (normal), epileptic, and post-treatment. A Leaky Integrate-and-Fire (LIF) neuron model is defined, as well as some key parameters (input current, synaptic gain, and refractory period) to reproduce the hallmark characteristics of epileptic activity, including increased firing rates, increased firing synchrony and population bursts.

The design is purposely simple: the aim is to provide a biologically accurate baseline, which allows high flexibility in parameters testing, and the possibility of running multiple trials to explore various network configurations.

Specifically, the model provides a foundation for testing hypotheses about seizure dynamics and intervention mechanisms. It may serve as a starting point for more advanced simulations in neuroscience research, offering insights into how local changes in connectivity and excitability can influence global network behaviour.

## Mathematical Framework

At the core of this simulation is the *Leaky Integrate-and-Fire (LIF)* neuron model, which is a simple but effective way to mimic neurons behaviour. The underlying idea is that neurons slowly build up electrical charge when they receive input, but they also “leak” that charge over time. If the charge is high enough, the neuron fires a spike and resets, just like a real neuron sending an electrical signal.

Several biological features are represented in a simplified form:

- *Synaptic input summation*: Each neuron can receive input from others, and these inputs are summed together to determine how much a neuron's voltage increases.
- *Noise injection*: Small random fluctuations are added to the input to simulate more natural variability.
- *Refractory periods*: After firing, each neuron “rests” for a short time before it can fire again, which helps avoid unrealistically non-stopping fast spiking.

The tools through which the system's behaviour was analysed are:

- *Firing rate calculations*: help measure how active the network is under each condition, giving a quick summary of how much the neurons are firing.
- *Spike correlations*: are used to look at synchrony, that is how much the neurons tend to fire together. This is especially useful for identifying patterns linked to seizures, which are known to involve overly synchronised activity.
- *Burst detection*: involves splitting time into small windows (bins) and checking when the total spike activity suddenly jumps. These bursts are key indicators of pathological activity like seizures.

Altogether, these mathematical tools provide a simplified yet powerful way to explore how small changes in a neuron's input or network wiring can lead to big changes in brain-like behaviour for individuals who suffer from epilepsy.

## Structure of the Project

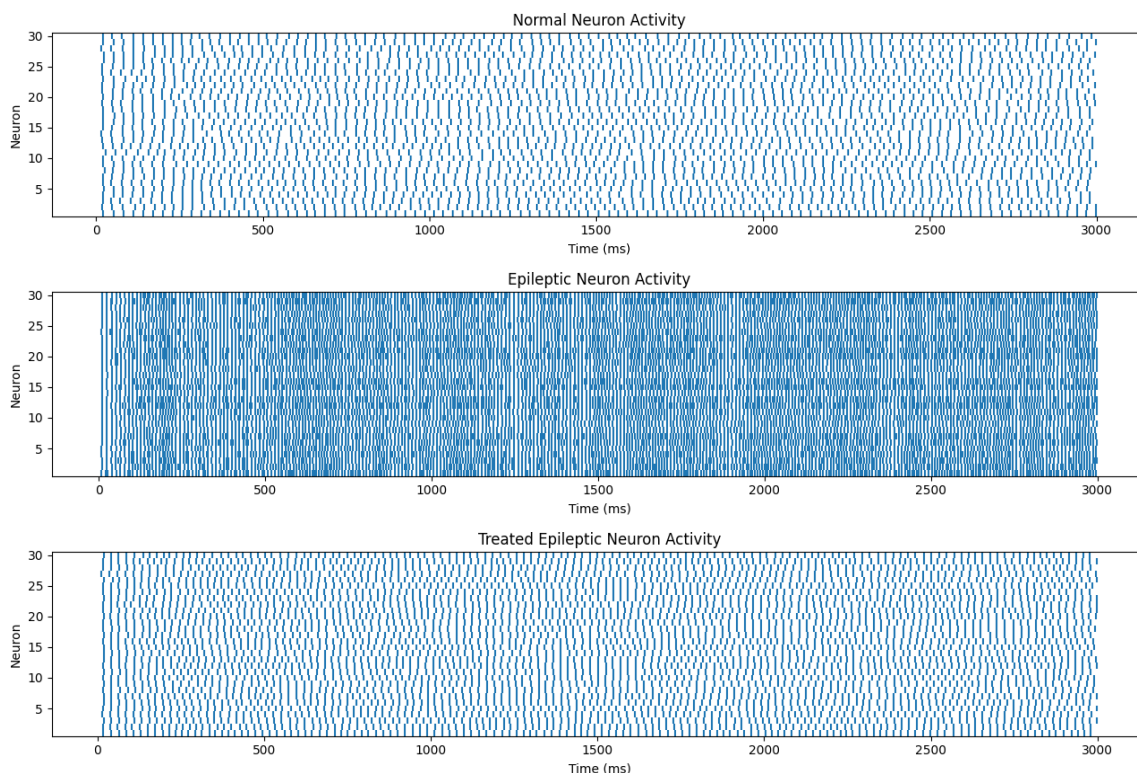
The network consists of 30 neurons simulated over 3000 milliseconds with a 1 ms time step. Parameters are chosen to balance both biological realism and computational simplicity, allowing for rapid testing and easy modification. The structure of the simulation is outlined below.

- **Parameter Setup:** I defined some fixed parameters; namely: duration, number of neurons, time step, and LIF-specific constants (resting potential, threshold, refractory periods). Then I focused specifically on the different conditions and defined their inputs and connectivity weights.
- **LIF Simulation function:** This function simulates spiking activity over time for each neuron.  
Features:
  - **Customisability:** Allows overriding membrane properties like to simulate altered cellular states.
  - **Noise Injection:** Simulates physiological variability by adding Gaussian noise to the input.
  - **Synaptic Connectivity:** Recurrent input from other neurons is included via a weighted connectivity matrix.
  - **Synaptic Gain Parameter:** Controls the impact of recurrent input.
- **Firing Rate Summary:** I computed the average firing rate across all neurons in each condition.
- **Correlation Analysis:** I computed a correlation matrix and average correlation from binned spike trains, to compare synchrony between neurons and among the different conditions.
- **Burst Detection and Population Dynamics:** I computed total network activity in short bins and highlighted time windows that exceed a set burst threshold (customisable) in the epileptic condition.

## Results

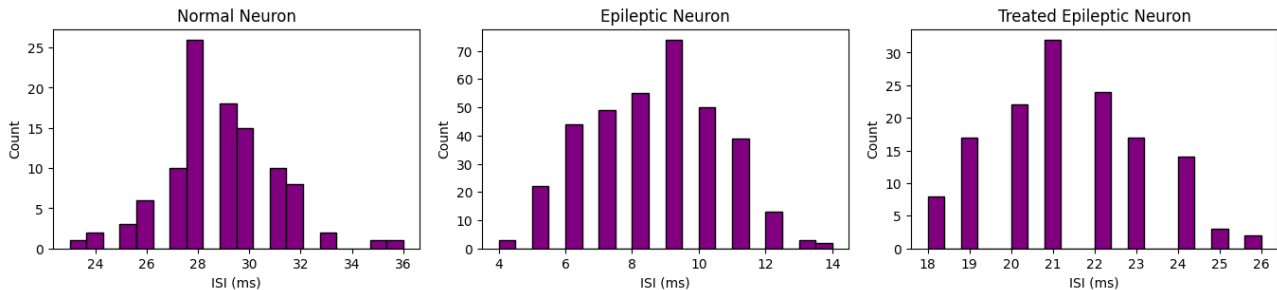
**Raster Plots:** These visualisations display spike timing for all 30 neurons over a 3000 ms window.

- **Normal:** Activity appears regular and desynchronised, with consistent firing across neurons; that's the general state of a healthy individual.
- **Epileptic:** Neurons exhibit dense and highly synchronous spiking, indicative of pathological over-excitation and loss of variability.
- **Treated:** Firing is visibly reduced and more regular, although still denser than that of a healthy individual, resembling the normal condition and suggesting a partial restoration of balance.



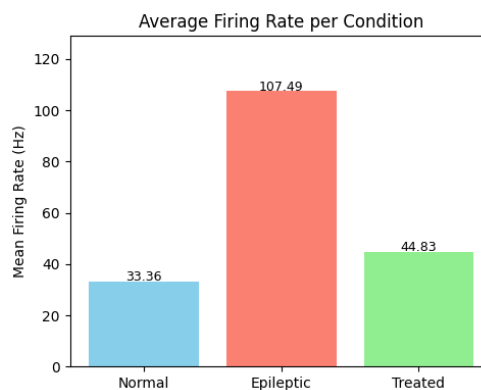
**ISI Histograms:** These plots show the Inter-Spike Interval (ISI) distribution for a representative neuron in each condition.

- Normal: ISIs are tightly distributed around ~28 ms, reflecting stable, more regular firing.
- Epileptic: ISIs shift towards shorter intervals (~9 ms), showing rapid, high-frequency firing typical of seizures.
- Treated: ISIs return to intermediate values (~21 ms), suggesting reduced excitability and a more controlled firing pattern.

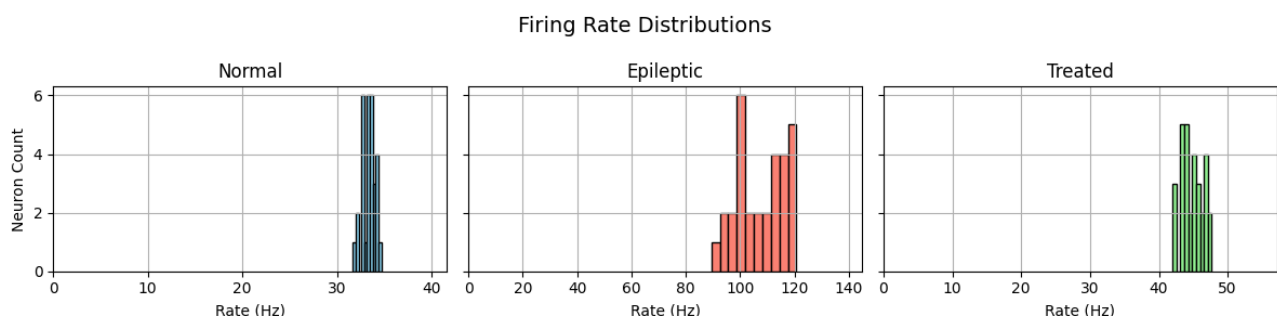


**Firing Rate Analysis:** The two visualisations below provide a clear summary of firing rate patterns across the three simulated conditions.

The bar plot shows that the epileptic network has a dramatically higher mean firing rate (~107 Hz), compared to the normal (~33 Hz) and treated (~45 Hz) cases. This aligns with the expected hyper-excitability characteristic of epilepsy displayed above.

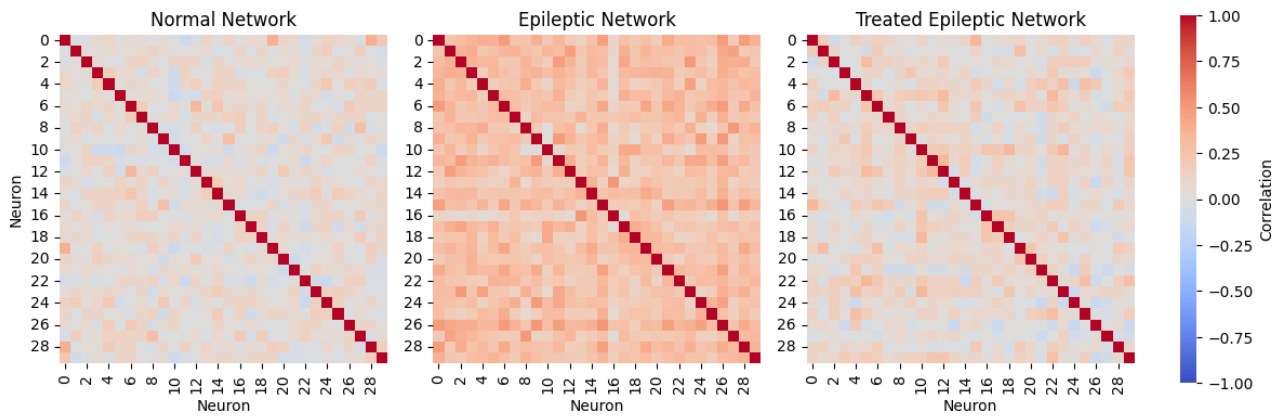


The plots below, show the distributions of firing rates per neuron, and they confirm once again that epileptic neurons are firing more rapidly on average, but also that they are more variable. In contrast, the normal network has a tight distribution centred around lower rates, while the treated network shows reduced rates compared to the epileptic case, though slightly higher than normal. This suggests partial restoration of more regular activity through treatment.



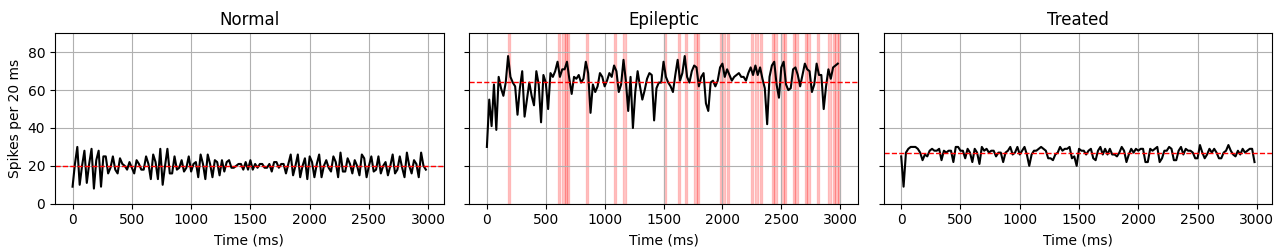
**Functional Connectivity via Spike Correlation:** The heatmaps resulting from the study of pairwise spike correlations are symmetric, and they show a clear difference in synchrony between the three conditions.

The epileptic network exhibits a noticeably higher average pairwise correlation among neurons ( $\approx 0.27$ ), indicative of increased synchrony and potential pathological bursting. In contrast, the normal network shows low average correlation ( $\approx 0.05$ ), reflecting desynchronised firing, while the treated network presents a mild elevation ( $\approx 0.07$ ), suggesting partial suppression of pathological synchrony.



**Population Activity and Burst Detection:** These last graphs illustrate the temporal evolution of population activity across the three conditions. In the epileptic case, bursts of high activity are clearly detected (highlighted in red), reflecting abnormal, excessive synchrony typical of seizure-like dynamics. The normal network shows stable, low-amplitude fluctuations around the average activity level, while the treated condition significantly reduces the number and amplitude of bursts, restoring dynamics closer to the baseline.

Population Activity Across Conditions (Bursts Highlighted)



## Limitations and Improvements

This project uses a simplified LIF model, without incorporating inhibitory neurons, advanced synaptic dynamics, spatial structure or specific pharmacological characteristics of treatment. These limitations certainly reduce biological realism but allow for clarity and baseline fast experimentation of common hyperparameters.

Future improvements could include modelling inhibitory circuits (e.g., GABAergic neurons), plasticity mechanisms and more detailed neuron models (heterogeneous thresholds, membrane properties, input currents...) to better capture the complexity of neuron networks, epileptic dynamics and treatments.